THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. To find whether the volume of water affects its boiling time, a student boils 100 mL in a steel pot on a strong flame and 200 mL in a glass beaker on a weak flame. Why is this test unfair? 1

(a) the water is heated at all
(b) the volumes of water are different
(c) time is measured with a clock
(d) the container and the flame also differ along with the volume

2. A scientist finds a plant with parallel veins, fibrous roots and a soft stem. Which is most probable? 1

(a) a peepal tree **(b)** a mango tree **(c)** a rose bush **(d)** a grass

3. A food gives a blue-black colour with iodine and an oily patch on paper. It contains: 1

(a) starch and fat **(b)** fat only
(c) starch only **(d)** protein and fibre

4. Three identical-looking bars P, Q, R: P attracts Q at both ends, Q attracts R at both ends, R repels P at one end. Which can be concluded? 1

(a) all are iron
(b) all are plastic
(c) all are magnets
(d) P and R are magnets, Q is magnetic material

5. A rope is 3.2 m long. After cutting off pieces of 45 cm, 55 cm and 60 cm, the remaining length is: 1

(a) 320 cm **(b)** 160 cm **(c)** 100 cm **(d)** 200 cm

6. A student heats the ends of four rods P, Q, R, S, each with a wax-fixed pin at the same distance. The pins fall at 10 s, 25 s, 40 s and 60 s. The best conductor is: 1

(a) P **(b)** S **(c)** Q **(d)** R

7. Thermometer A reads 0 °C in melting ice and 100 °C in boiling water. Thermometer B reads 2 °C and 98 °C. For the same hot water at actual 50 °C, B would most likely read: 1

(a) 54 °C **(b)** 50 °C **(c)** 48 °C **(d)** 52 °C

8. When ice at 0 °C is heated, the temperature stays at 0 °C till all ice melts. This shows that: 1

(a) the heater is weak **(b)** heat is used to change the state
(c) the thermometer is faulty **(d)** ice cannot be heated

9. A mixture of iron filings, common salt, and fine sand is given. Which sequence is the most efficient? 1

(a) magnet, add water and filter, evaporate filtrate
(b) filter, magnet, sieve
(c) evaporate first, filter, magnet
(d) add water, evaporate, magnet

10. A candle burns, melts, gives out waste gases and its flame responds to wind. Why is it still non-living? 1

(a) it is hot
(b) it makes light
(c) it melts
(d) it does not grow or reproduce on its own

11. A student collected 10 mL of gas from a jar over water after a candle burnt out. The candle extinguished because: 1

(a) all nitrogen was used
(b) the jar broke
(c) no gas remained
(d) most oxygen was used

12. If sunlight takes 8 minutes to reach the Earth, the Sun's distance in terms of light travel time is: 1

(a) 8 light-hours **(b)** 8 light-years **(c)** 8 light-seconds **(d)** 8 light-minutes

13. A study found that more ice cream is sold when more people get sunburnt. The best explanation is: 1

(a) both depend on hot weather
(b) ice cream causes sunburn
(c) it is magic
(d) sunburn causes ice cream sales

14. Which of the following is a correct inference about a tiger's eyes in front of the face? 1

(a) helps fly
(b) helps judge distance for hunting
(c) helps to swim
(d) helps see all sides

15. Why do nutritionists advise limiting packaged drinks? 1

(a) they have fibre
(b) they have vitamins
(c) they contain protein
(d) they contain much sugar and few nutrients

16. Which of the following is the best argument for using a magnet to separate iron from copper turnings? 1

(a) iron is magnetic and copper is not
(b) both are shiny
(c) both are metals
(d) both rust

17. **Assertion (A):** A hypothesis is a guess that can be tested. 1
Reason (R): An experiment can show whether a hypothesis is supported or not.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Wheat plants have fibrous roots. 1
Reason (R): Plants with fibrous roots generally have leaves with parallel venation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** Proteins are called body-building nutrients. 1
Reason (R): Proteins help in growth and repair of the body.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** A freely suspended bar magnet always comes to rest along the north-south direction. 1
Reason (R): The Earth behaves like a huge magnet.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. In an experiment on plant growth, A is kept in sunlight and B in shade, but A is also given more water. Why is this unfair? 2

22. Cactus leaves are spines but it still makes food. How? 2

23. A food sample turns blue-black with iodine and leaves an oily spot on paper. What conclusions can you draw? 2

24. A compass is placed on a steel table and gives a wrong reading. Explain and suggest a remedy. 2

25. A road runs 3.6 km. A boy walks at a step of 60 cm. How many steps does he take? 2

26. Explain why an empty plastic bottle floats but a bottle filled with sand sinks. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Design a fair test to compare which of three fabrics dries the fastest. Name controlled variables. 3

28. A plant has leaves with net veins, a tap root, a thick woody trunk 15 m tall. Classify it and give the chief features of each part. 3

29. A girl's diet: breakfast - bread and jam, lunch - noodles, dinner - fried snacks. Evaluate it and suggest changes. 3

30. Describe how to show that the poles of a magnet are not at the very ends but near them, using iron filings. 3

31. The circumference of a wheel is 2.2 m. How many turns will it make to cover 1.1 km? 3

OR

Substance X is a shiny solid that does not dissolve in water, sinks in water and becomes hot at the other end when one end is heated. Name its class of material and give two reasons. 3

32. A student measures the temperature of boiling water as 98 °C in a hill station and 100 °C in a plain. Suggest a reason. 3

OR

A tank of 2000 L loses 4 L per day by evaporation. If it is filled once a month (30 days) how much is lost, and what percentage? 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Riya noticed that the plant on her balcony grew taller than the plant kept inside her cupboard. She guessed that plants need sunlight to grow well. She took two identical pots with the same soil and gave both the same amount of water. She kept one in sunlight and the other in a dark cupboard, and noted the height of both plants every two days for ten days. 5
(i) What is Riya's guess called?
(ii) Which one thing was different for the two pots?
(iii) Name the step in which she noted the heights every two days.
(iv) Why did she keep the soil and water the same? What is the likely conclusion?

34. **Case study.** A class visited a garden in Jaipur and listed these plants: tulsi, rose, mango, pumpkin and grapevine. The teacher said plants can be grouped as herbs, shrubs, trees, creepers and climbers by the nature of their stems. 5
(i) Name the plant from the list that is a tree.
(ii) Which plant is a creeper?
(iii) Give one difference between a herb and a shrub.
(iv) How is a climber different from a creeper? Give an example of a climber from the list.

35. A group of students are tested for health. Student A has bleeding gums, B has night blindness, C has swollen neck, D has bent legs, E is pale. Identify the cause, the remedy for each. 5

36. Design an investigation to compare the strength of two magnets using paper clips and describe how to make it a fair test. 5

OR

A student wants to find the circumference of a tree trunk, the thickness of a coin and the length of a pencil. Choose the instruments and explain the method for each. 5

37. Design an investigation to rank wood, plastic, steel and copper for suitability as a cooking spoon handle. Name the variables and the safety steps. 5

OR

Describe an experiment to find which of three liquids (water, oil, milk) cools fastest. Explain controls, measurements and how to present results. 5

38. Design an experiment to find which of the three places (sunny, shady, windy) is best for drying clothes and how to compare. 5

OR

Two beakers contain salt water and muddy water. Explain with the help of experiments how you would show that one has dissolved salt and the other has insoluble mud. 5

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND